4.3 Connections with Cohomology

The Hurewicz theorem provides a strong link between homotopy groups and homology, and hence also an indirect relation with cohomology. But there is a more direct connection with cohomology of a quite different sort. We will show that for every CW complex X there is a natural bijection between $H^n(X;G)$ and the set $\langle X, K(G,n)\rangle$ of basepoint-preserving homotopy classes of maps from X to a $K(G,n)$. We will also define a natural group structure on $\langle X, K(G,n)\rangle$ that makes the bijection a group isomorphism. The mere fact that there is any connection at all between cohomology and homotopy classes of maps is the first surprise here, and the second is that Eilenberg–MacLane spaces are involved, since their definition is entirely in terms of homotopy groups, which on the face of it have nothing to do with cohomology.

After proving this basic isomorphism $H^n(X;G) \approx \langle X, K(G,n)\rangle$ and describing a few of its immediate applications, the later parts of this section aim toward a further study of Postnikov towers, which were introduced briefly in §4.1. These provide a general theoretical method for realizing an arbitrary CW complex as a sort of twisted product of Eilenberg–MacLane spaces, up to homotopy equivalence. The most geometric interpretation of the phrase 'twisted product' is the notion of fiber bundle introduced in the previous section, but here we need the more homotopy-theoretic notion of a *fibration*, so before we begin the discussion of Postnikov towers we first take a few pages to present some basic constructions and results about fibrations.

As we shall see, Postnikov towers can be expressed as sequences of fibrations with fibers Eilenberg–MacLane spaces, so we can again expect close connections with cohomology. One such connection is provided by *k-invariants*, which describe, at least in principle, how Postnikov towers for a broad class of spaces are determined by a sequence of cohomology classes. Another application of these ideas, described at the end of the section, is a technique for factoring basic extension and lifting problems in homotopy theory into a sequence of smaller problems whose solutions are equivalent to the vanishing of certain cohomology classes. This technique goes under the somewhat grandiose title of Obstruction Theory, though it is really quite a simple idea when expressed in terms of Postnikov towers.

The Homotopy Construction of Cohomology

The main result of this subsection is the following fundamental relationship between singular cohomology and Eilenberg–MacLane spaces:

Theorem 4.57. *There are natural bijections $T : \langle X, K(G,n)\rangle \to H^n(X;G)$ for all CW complexes X and all $n > 0$, with G any abelian group. Such a T has the form $T([f]) = f^*(\alpha)$ for a certain distinguished class $\alpha \in H^n(K(G,n);G)$.*

In the course of the proof we will define a natural group structure on $\langle X, K(G,n)\rangle$ such that the transformation T is an isomorphism.

A class $\alpha \in H^n(K(G,n);G)$ with the property stated in the theorem is called a **fundamental class**. The proof of the theorem will yield an explicit fundamental class, namely the element of $H^n(K(G,n);G) \approx \mathrm{Hom}(H_n(K;\mathbb{Z}),G)$ given by the inverse of the Hurewicz isomorphism $G = \pi_n(K(G,n)) \to H_n(K;\mathbb{Z})$. Concretely, if we choose $K(G,n)$ to be a CW complex with $(n-1)$-skeleton a point, then a fundamental class is represented by the cellular cochain assigning to each n-cell of $K(G,n)$ the element of $\pi_n(K(G,n))$ defined by a characteristic map for the n-cell.

For connected CW complexes X the theorem also holds with $\langle X, K(G,n)\rangle$ replaced by $[X, K(G,n)]$, the nonbasepointed homotopy classes. This is easy to see when $n > 1$ since every map $X \to K(G,n)$ can be homotoped to take basepoint to basepoint, and every homotopy between basepoint-preserving maps can be homotoped to be basepoint-preserving if the target space $K(G,n)$ is simply-connected. When $n = 1$ the equality $[X, K(G,n)] = \langle X, K(G,n)\rangle$ is an exercise for §4.A, using the assumption that G is abelian.

It is possible to give a direct, bare-hands proof of the theorem, constructing maps and homotopies cell by cell. This provides much geometric insight into why the result is true, but unfortunately the technical details of this proof are rather tedious. So we shall take a different approach, one that has the advantage of placing the result in its natural context via general machinery that turns out to be quite useful in other situations as well. The two main steps will be the following assertions.

(1) The functors $h^n(X) = \langle X, K(G,n)\rangle$ define a reduced cohomology theory on the category of basepointed CW complexes.

(2) If a reduced cohomology theory h^* defined on CW complexes has coefficient groups $h^n(S^0)$ which are zero for $n \neq 0$, then there are natural isomorphisms $h^n(X) \approx \widetilde{H}^n(X; h^0(S^0))$ for all CW complexes X and all n.

Towards proving (1) we will study a more general question: When does a sequence of spaces K_n define a cohomology theory by setting $h^n(X) = \langle X, K_n\rangle$? Note that this will be a reduced cohomology theory since $\langle X, K_n\rangle$ is trivial when X is a point.

The first question to address is putting a group structure on the set $\langle X, K\rangle$. This requires that either X or K have some special structure. When $X = S^n$ we have $\langle S^n, K\rangle = \pi_n(K)$, which has a group structure when $n > 0$. The definition of this group structure works more generally whenever S^n is replaced by a suspension SX, with the sum of maps $f, g : SX \to K$ defined as the composition $SX \to SX \vee SX \to K$ where the first map collapses an 'equatorial' $X \subset SX$ to a point and the second map consists of f and g on the two summands. However, for this to make sense we must be talking about basepoint-preserving maps, and there is a problem with where to choose the basepoint in SX. If x_0 is a basepoint of X, the basepoint of SX should be somewhere along the segment $\{x_0\} \times I \subset SX$, most likely either an endpoint or the midpoint, but no single choice of such a basepoint gives a well-defined sum. The sum would be well-defined if we restricted attention to maps sending the whole segment $\{x_0\}\times I$ to the basepoint. This is equivalent to considering basepoint-preserving maps $\Sigma X \to K$ where $\Sigma X = SX/(\{x_0\}\times I)$ and the image of $\{x_0\}\times I$ in ΣX is taken to be the basepoint. If X is a CW complex with x_0 a 0-cell, the quotient map $SX \to \Sigma X$ is a homotopy equivalence since it collapses a contractible subcomplex of SX to a point, so we can identify $\langle SX, K\rangle$ with $\langle \Sigma X, K\rangle$. The space ΣX is called the **reduced suspension** of X when we want to distinguish it from the ordinary suspension SX.

It is easy to check that $\langle \Sigma X, K\rangle$ is a group with respect to the sum defined above, inverses being obtained by reflecting the I coordinate in the suspension. However, what we would really like to have is a group structure on $\langle X, K\rangle$ arising from a special structure on K rather than on X. This can be obtained using the following basic **adjoint relation**:

- $\langle \Sigma X, K\rangle = \langle X, \Omega K\rangle$ where ΩK is the space of loops in K at its chosen basepoint and the constant loop is taken as the basepoint of ΩK.

The space ΩK, called the **loopspace** of K, is topologized as a subspace of the space K^I of all maps $I \to K$, where K^I is given the compact-open topology; see the Appendix for the definition and basic properties of this topology. The adjoint relation $\langle \Sigma X, K\rangle = \langle X, \Omega K\rangle$ holds because basepoint-preserving maps $\Sigma X \to K$ are exactly the same as basepoint-preserving maps $X \to \Omega K$, the correspondence being given by associating to $f : \Sigma X \to K$ the family of loops obtained by restricting f to the images of the segments $\{x\}\times I$ in ΣX.

Taking $X = S^n$ in the adjoint relation, we see that $\pi_{n+1}(K) = \pi_n(\Omega K)$ for all $n \geq 0$. Thus passing from a space to its loopspace has the effect of shifting homotopy groups down a dimension. In particular we see that $\Omega K(G,n)$ is a $K(G,n-1)$. This fact will turn out to be important in what follows.

Note that the association $X \mapsto \Omega X$ is a functor: A basepoint-preserving map $f : X \to Y$ induces a map $\Omega f : \Omega X \to \Omega Y$ by composition with f. A homotopy $f \simeq g$ induces a homotopy $\Omega f \simeq \Omega g$, so it follows formally that $X \simeq Y$ implies $\Omega X \simeq \Omega Y$.

It is a theorem of [Milnor 1959] that the loopspace of a CW complex has the homotopy type of a CW complex. This may be a bit surprising since loopspaces are usually quite large spaces, though of course CW complexes can be quite large too, in terms of the number of cells. What often happens in practice is that if a CW complex X has only finitely many cells in each dimension, then ΩX is homotopy equivalent to a CW complex with the same property. We will see explicitly how this happens for $X = S^n$ in §4.J.

Composition of loops defines a map $\Omega K \times \Omega K \to \Omega K$, and this gives a sum operation in $\langle X, \Omega K\rangle$ by setting $(f+g)(x) = f(x) \cdot g(x)$, the composition of the loops $f(x)$ and $g(x)$. Under the adjoint relation this is the same as the sum in $\langle \Sigma X, K\rangle$ defined previously. If we take the composition of loops as the sum operation then it is perhaps somewhat easier to see that $\langle X, \Omega K\rangle$ is a group since the same reasoning which shows that $\pi_1(K)$ is a group can be applied.

Since cohomology groups are abelian, we would like the group $\langle X, \Omega K\rangle$ to be abelian. This can be achieved by iterating the operation of forming loopspaces. One has a double loopspace $\Omega^2 K = \Omega(\Omega K)$ and inductively an n-fold loopspace $\Omega^n K = \Omega(\Omega^{n-1}K)$. The evident bijection $K^{Y\times Z} \approx (K^Y)^Z$ is a homeomorphism for locally compact Hausdorff spaces Y and Z, as shown in Proposition A.16 in the Appendix, and from this it follows by induction that $\Omega^n K$ can be regarded as the space of maps $I^n \to K$ sending ∂I^n to the basepoint. Taking $n = 2$, we see that the argument that $\pi_2(K)$ is abelian shows more generally that $\langle X, \Omega^2 K\rangle$ is an abelian group. Iterating the adjoint relation gives $\langle \Sigma^n X, K\rangle = \langle X, \Omega^n K\rangle$, so this is an abelian group for all $n \geq 2$.

Thus for a sequence of spaces K_n to define a cohomology theory $h^n(X) = \langle X, K_n\rangle$ we have been led to the assumption that each K_n should be a loopspace and in fact a double loopspace. Actually we do not need K_n to be literally a loopspace since it would suffice for it to be homotopy equivalent to a loopspace, as $\langle X, K_n\rangle$ depends only on the homotopy type of K_n. In fact it would suffice to have just a weak homotopy equivalence $K_n \to \Omega L_n$ for some space L_n since this would induce a bijection $\langle X, K_n\rangle = \langle X, \Omega L_n\rangle$ by Proposition 4.22. In the special case that $K_n = K(G,n)$ for all n, we can take $L_n = K_{n+1} = K(G, n+1)$ by the earlier observation that $\Omega K(G, n+1)$ is a $K(G,n)$. Thus if we take the $K(G,n)$'s to be CW complexes, the map $K_n \to \Omega K_{n+1}$ is just a CW approximation $K(G,n) \to \Omega K(G, n+1)$.

There is another reason to look for weak homotopy equivalences $K_n \to \Omega K_{n+1}$. For a reduced cohomology theory $h^n(X)$ there are natural isomorphisms $h^n(X) \approx h^{n+1}(\Sigma X)$ coming from the long exact sequence of the pair (CX, X) with CX the cone on X, so if $h^n(X) = \langle X, K_n\rangle$ for all n then the isomorphism $h^n(X) \approx h^{n+1}(\Sigma X)$ translates into a bijection $\langle X, K_n\rangle \approx \langle \Sigma X, K_{n+1}\rangle = \langle X, \Omega K_{n+1}\rangle$ and the most natural thing would be for this to come from a weak equivalence $K_n \to \Omega K_{n+1}$. Weak equivalences of this form would give also weak equivalences $K_n \to \Omega K_{n+1} \to \Omega^2 K_{n+2}$ and so we would automatically obtain an abelian group structure on $\langle X, K_n\rangle \approx \langle X, \Omega^2 K_{n+2}\rangle$.

These observations lead to the following definition. An **Ω-spectrum** is a sequence of CW complexes $K_1, K_2, \cdots$ together with weak homotopy equivalences $K_n \to \Omega K_{n+1}$ for all n. By using the theorem of Milnor mentioned above it would be possible to replace 'weak homotopy equivalence' by 'homotopy equivalence' in this definition. However it does not noticeably simplify matters to do this, except perhaps psychologically.

Notice that if we discard a finite number of spaces K_n from the beginning of an Ω-spectrum $K_1, K_2, \cdots$, then these omitted terms can be reconstructed from the remaining K_n's since each K_n determines K_{n-1} as a CW approximation to ΩK_n. So it is not important that the sequence start with K_1. By the same token, this allows us to extend the sequence of K_n's to all negative values of n. This is significant because a general cohomology theory $h^n(X)$ need not vanish for negative n.

Theorem 4.58. *If $\{K_n\}$ is an Ω-spectrum, then the functors $X \mapsto h^n(X) = \langle X, K_n \rangle$, $n \in \mathbb{Z}$, define a reduced cohomology theory on the category of basepointed CW complexes and basepoint-preserving maps.*

Rather amazingly, the converse is also true: Every reduced cohomology theory on CW complexes arises from an Ω-spectrum in this way. This is the Brown representability theorem which will be proved in §4.E.

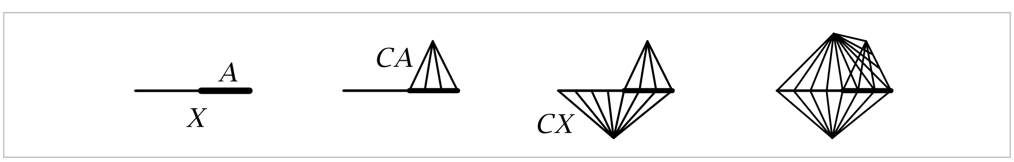
A space K_n in an Ω-spectrum is sometimes called an *infinite loopspace* since there are weak homotopy equivalences $K_n \to \Omega^k K_{n+k}$ for all k. A number of important spaces in algebraic topology turn out to be infinite loopspaces. Besides Eilenberg–MacLane spaces, two other examples are the infinite-dimensional orthogonal and unitary groups O and U, for which there are weak homotopy equivalences $O \to \Omega^8 O$ and $U \to \Omega^2 U$ by a strong form of the Bott periodicity theorem, as we will show in [VBKT]. So O and U give periodic Ω-spectra, hence periodic cohomology theories known as real and complex K–theory. For a more in-depth introduction to the theory of infinite loopspaces, the book [Adams 1978] can be much recommended.

Proof: Two of the three axioms for a cohomology theory, the homotopy axiom and the wedge sum axiom, are quite easy to check. For the homotopy axiom, a basepoint-preserving map $f : X \to Y$ induces $f^* : \langle Y, K_n \rangle \to \langle X, K_n \rangle$ by composition, sending a map $Y \to K_n$ to $X \xrightarrow{f} Y \to K_n$. Clearly f^* depends only on the basepoint-preserving homotopy class of f, and it is obvious that f^* is a homomorphism if we replace K_n by ΩK_{n+1} and use the composition of loops to define the group structure. The wedge sum axiom holds since in the realm of basepoint-preserving maps, a map $\bigvee_\alpha X_\alpha \to K_n$ is the same as a collection of maps $X_\alpha \to K_n$.

The bulk of the proof involves associating a long exact sequence to each CW pair (X, A). As a first step we build the following diagram:

$$\begin{tikzcd}
A \arrow[r, hook] \arrow[d, equal] & X \arrow[r, hook] \arrow[d, equal] & X \cup CA \arrow[r, hook] \arrow[d, "\simeq"'] & (X \cup CA) \cup CX \arrow[r, hook] \arrow[d, shift right, "\simeq"'] & \big((X \cup CA) \cup CX\big) \cup C(X \cup CA) \arrow[d, shift right, "\simeq"'] \\
A \arrow[r, hook] & X \arrow[r] & X/A \arrow[r] & SA \arrow[r, "\subset"] \arrow[u, shift right] & SX \arrow[u, shift right]
\end{tikzcd} \tag{1}$$

The first row is obtained from the inclusion $A \hookrightarrow X$ by iterating the rule, 'attach a cone on the preceding subspace,' as shown in the pictures below.



The three downward arrows in the diagram (1) are quotient maps collapsing the most recently attached cone to a point. Since cones are contractible, these downward maps are homotopy equivalences. The second and third of them have homotopy inverses the evident inclusion maps, indicated by the upward arrows. In the lower row of the diagram the maps are the obvious ones, except for the map $X/A \to SX$ which is the composition of a homotopy inverse of the quotient map $X \cup CA \to X/A$ followed by the maps $X \cup CA \to (X \cup CA) \cup CX \to SA$. Thus the square containing this map commutes up to homotopy. It is easy to check that the same is true of the right-hand square as well.

The whole construction can now be repeated with $SA \hookrightarrow SX$ in place of $A \hookrightarrow X$, then with double suspensions, and so on. The resulting infinite sequence can be written in either of the following two forms:

$$A \to X \to X \cup CA \to SA \to SX \to S(X \cup CA) \to S^2A \to S^2X \to \cdots$$
$$A \to X \to X/A \to SA \to SX \to SX/SA \to S^2A \to S^2X \to \cdots$$

In the first version we use the obvious equality $SX \cup CSA = S(X \cup CA)$. The first version has the advantage that the map $X \cup CA \to SA$ is easily described and canonical, whereas in the second version the corresponding map $X/A \to SA$ is only defined up to homotopy since it depends on choosing a homotopy inverse to the quotient map $X \cup CA \to X/A$. The second version does have the advantage of conciseness, however.

When basepoints are important it is generally more convenient to use reduced cones and reduced suspensions, obtained from ordinary cones and suspensions by collapsing the segment $\{x_0\} \times I$ where x_0 is the basepoint. The image point of this segment in the reduced cone or suspension then serves as a natural basepoint in the quotient. Assuming x_0 is a 0-cell, these collapses of $\{x_0\} \times I$ are homotopy equivalences. Using reduced cones and suspensions in the preceding construction yields a sequence

$$(2) \qquad A \hookrightarrow X \to X/A \to \Sigma A \hookrightarrow \Sigma X \to \Sigma(X/A) \to \Sigma^2 A \hookrightarrow \Sigma^2 X \to \cdots$$

where we identify $\Sigma X/\Sigma A$ with $\Sigma(X/A)$, and all the later maps in the sequence are suspensions of the first three maps. This sequence, or its unreduced version, is called the **cofibration sequence** or **Puppe sequence** of the pair (X, A). It has an evident naturality property, namely, a map $(X, A) \to (Y, B)$ induces a map between the cofibration sequences of these two pairs, with homotopy-commutative squares:

$$\begin{array}{ccccccccccccccc}
A & \longrightarrow & X & \longrightarrow & X/A & \longrightarrow & \Sigma A & \longrightarrow & \Sigma X & \longrightarrow & \Sigma(X/A) & \longrightarrow & \Sigma^2 A & \longrightarrow & \cdots \\
\downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\
B & \longrightarrow & Y & \longrightarrow & Y/B & \longrightarrow & \Sigma B & \longrightarrow & \Sigma Y & \longrightarrow & \Sigma(Y/B) & \longrightarrow & \Sigma^2 B & \longrightarrow & \cdots
\end{array}$$

Taking basepoint-preserving homotopy classes of maps from the spaces in (2) to a fixed space K gives a sequence

$$(3) \qquad \langle A, K\rangle \leftarrow \langle X, K\rangle \leftarrow \langle X/A, K\rangle \leftarrow \langle \Sigma A, K\rangle \leftarrow \langle \Sigma X, K\rangle \leftarrow \cdots$$

whose maps are defined by composition with those in (2). For example, the map $\langle X, K\rangle \to \langle A, K\rangle$ sends a map $X \to K$ to $A \to X \to K$. The sets in (3) are groups starting with $\langle \Sigma A, K\rangle$, and abelian groups from $\langle \Sigma^2 A, K\rangle$ onward. It is easy to see that the maps between these groups are homomorphisms since the maps in (2) are suspensions from $\Sigma A \to \Sigma X$ onward. In general the first three terms of (3) are only sets with distinguished 'zero' elements, the constant maps.

A key observation is that the sequence (3) is exact. To see this, note first that the diagram (1) shows that, up to homotopy equivalence, each term in (2) is obtained from its two predecessors by the same procedure of forming a mapping cone, so it suffices to show that $\langle A, K\rangle \leftarrow \langle X, K\rangle \leftarrow \langle X \cup CA, K\rangle$ is exact. This is easy: A map $f : X \to K$ goes to zero in $\langle A, K\rangle$ iff its restriction to A is nullhomotopic, fixing the basepoint, and this is equivalent to f extending to a map $X \cup CA \to K$.

If we have a weak homotopy equivalence $K \to \Omega K'$ for some space K', then the sequence (3) can be continued three steps to the left via the commutative diagram

$$\begin{tikzcd}
 & & & \langle A,K\rangle \arrow[d, "\approx"] & \langle X,K\rangle \arrow[l] \arrow[d, "\approx"] & \langle X/A,K\rangle \arrow[l] \arrow[d, "\approx"] & \cdots \arrow[l] \\
 & & & \langle A,\Omega K'\rangle \arrow[d, "\approx"] & \langle X,\Omega K'\rangle \arrow[l] \arrow[d, "\approx"] & \langle X/A,\Omega K'\rangle \arrow[l] \arrow[d, "\approx"] & \cdots \arrow[l] \\
\langle A,K'\rangle & \langle X,K'\rangle \arrow[l] & \langle X/A,K'\rangle \arrow[l] & \langle \Sigma A,K'\rangle \arrow[l] & \langle \Sigma X,K'\rangle \arrow[l] & \langle \Sigma(X/A),K'\rangle \arrow[l] & \cdots \arrow[l]
\end{tikzcd}$$

Thus if we have a sequence of spaces K_n together with weak homotopy equivalences $K_n \to \Omega K_{n+1}$, we can extend the sequence (3) to the left indefinitely, producing a long exact sequence

$$(4) \qquad \cdots \leftarrow \langle A, K_n\rangle \leftarrow \langle X, K_n\rangle \leftarrow \langle X/A, K_n\rangle \leftarrow \langle A, K_{n-1}\rangle \leftarrow \langle X, K_{n-1}\rangle \leftarrow \cdots$$

All the terms here are abelian groups and the maps homomorphisms. This long exact sequence is natural with respect to maps $(X,A) \to (Y,B)$ since cofibration sequences are natural. □

There is no essential difference between cohomology theories on basepointed CW complexes and cohomology theories on nonbasepointed CW complexes. Given a reduced basepointed cohomology theory $\widetilde{h}^*$, one gets an unreduced theory by setting $h^n(X,A) = \widetilde{h}^n(X/A)$, where $X/\varnothing = X_+$, the union of X with a disjoint basepoint. This is a nonbasepointed theory since an arbitrary map $X \to Y$ induces a basepoint-preserving map $X_+ \to Y_+$. Furthermore, a nonbasepointed unreduced theory h^* gives a nonbasepointed reduced theory by setting $\widetilde{h}^n(X) = \mathrm{Coker}(h^n(\mathit{point}) \to h^n(X))$, where the map is induced by the constant map $X \to \mathit{point}$. One could also give an argument using suspension, which is always an isomorphism for reduced theories, and which takes one from the nonbasepointed to the basepointed category.

Theorem 4.59. *If h^* is an unreduced cohomology theory on the category of CW pairs and $h^n(\mathit{point}) = 0$ for $n \neq 0$, then there are natural isomorphisms $h^n(X,A) \approx H^n(X,A;h^0(\mathit{point}))$ for all CW pairs (X,A) and all n. The corresponding statement for homology theories is also true.*

Proof: The case of homology is slightly simpler, so let us consider this first. For CW complexes, relative homology groups reduce to absolute groups, so it suffices to deal with the latter. For a CW complex X the long exact sequences of h_* homology groups for the pairs (X^n, X^{n-1}) give rise to a cellular chain complex

$$\cdots \longrightarrow h_{n+1}(X^{n+1}, X^n) \xrightarrow{d_{n+1}} h_n(X^n, X^{n-1}) \xrightarrow{d_n} h_{n-1}(X^{n-1}, X^{n-2}) \longrightarrow \cdots$$

just as for ordinary homology. The hypothesis that $h^n(point) = 0$ for $n \neq 0$ implies that this chain complex has homology groups $h_n(X)$ by the same argument as for ordinary homology. The main thing to verify now is that this cellular chain complex is isomorphic to the cellular chain complex in ordinary homology with coefficients in the group $G = h_0(point)$. Certainly the cellular chain groups in the two cases are isomorphic, being direct sums of copies of G with one copy for each cell, so we have only to check that the cellular boundary maps are the same.

It is not really necessary to treat the cellular boundary map d_1 from 1-chains to 0-chains since one can always pass from X to ΣX, suspension being a natural isomorphism in any homology theory, and the double suspension $\Sigma^2 X$ has no 1-cells.

The calculation of cellular boundary maps d_n for $n > 1$ in terms of degrees of certain maps between spheres works equally well for the homology theory h_*, where 'degree' now means degree with respect to the h_* theory, so what is needed is the fact that a map $S^n \to S^n$ of degree m in the usual sense induces multiplication by m on $h_n(S^n) \approx G$. This is obviously true for degrees 0 and 1, represented by a constant map and the identity map. Since $\pi_n(S^n) \approx \mathbb{Z}$, every map $S^n \to S^n$ is homotopic to some multiple of the identity, so the general case will follow if we know that degree in the h_* theory is additive with respect to the sum operation in $\pi_n(S^n)$. This is a special case of the following more general assertion:

Lemma 4.60. *If a functor h from basepointed CW complexes to abelian groups satisfies the homotopy and wedge axioms, then for any two basepoint-preserving maps $f, g : \Sigma X \to K$, we have $(f+g)_* = f_* + g_*$ if h is covariant and $(f+g)^* = f^* + g^*$ if h is contravariant.*

Proof: The map $f + g$ is the composition $\Sigma X \xrightarrow{c} \Sigma X \vee \Sigma X \xrightarrow{f \vee g} K$ where c is the quotient map collapsing an equatorial copy of X. In the covariant case consider the diagram at the right, where i_1 and i_2 are the inclusions $\Sigma X \hookrightarrow \Sigma X \vee \Sigma X$. Let $q_1, q_2 : \Sigma X \vee \Sigma X \longrightarrow \Sigma X$ be the quotient maps restricting to the identity on the summand indicated by the subscript and collapsing the other summand to a point.

$$\begin{tikzcd} h(\Sigma X) \arrow[r, "c_*"] & h(\Sigma X \vee \Sigma X) \arrow[r, "(f \vee g)_*"] & h(K) \\ & h(\Sigma X) \oplus h(\Sigma X) \arrow[u, "i_{1*} \oplus i_{2*}", "\approx"'] & \end{tikzcd}$$

Then $q_{1*} \oplus q_{2*}$ is an inverse to $i_{1*} \oplus i_{2*}$ since $q_j i_k$ is the identity map for $j = k$ and the constant map for $j \neq k$.

An element x in the left-hand group $h(\Sigma X)$ in the diagram is sent by the composition $(q_{1*} \oplus q_{2*})c_*$ to the element (x, x) in the lower group $h(\Sigma X) \oplus h(\Sigma X)$ since